Heterogeneous initial states: weak instruments and independent-record inference

S16. Heterogeneous initial states and independent weak instruments

Consider independent records $Y_r = x_{\theta_0,u_r} + \eta_r$ of a deterministic polynomial ODE. The initial states may form a deterministic triangular array; alternatively they may be iid draws from a compact set, with the following conditions imposed conditionally on them. The sensor covariance is C_ν. Each Hermite-corrected, exactly integrated weak moment obeys

$$E\{m_r(\theta_0, \nu_0) \mid u_r\} = 0, \qquad m_r(\theta, \nu) = b_r - G_r(\nu)\theta.$$

This identity follows by taking expectations of the corrected polynomial and applying the ODE weak identity separately to each trajectory. Its validity is uniform over allowed initial states; the distribution of u_r need not be estimated. The numerical version needs the quadrature condition below.

S16.1. Numbered assumptions and fixed-noise estimating equations

(A1) The coefficient vector has fixed dimension p and belongs to an open identifiable neighborhood. Trajectories and their first coefficient/initial-state sensitivities exist on $[0, T]$. Initial states lie in a compact set on which the selected moments have uniformly bounded moments of order $2 + \delta$ for some $\delta > 0$.

(A2) Records and any auxiliary randomizations are independent across r. Noise is Gaussian with known positive covariance C_{ν_0}. Weak tests and quadrature rules are fixed independently of each record's noise. The coefficient structure is supplied.

(A3) For every record there are two conditionally independent views Y_r^+ and Y_r^- with mean x_{θ_0,u_r} and covariance $2C_{\nu_0}$. An instrument $L_r^\pm$ is measurable with respect to its own view and independent auxiliary training data. Corrected moments $m_r^\pm$ use the variance of their own view. The instrument map is fixed conditional on that training data and has the moment bounds required in (A1).

(A4) Put $B_r = (L_r^+ G_r^- + L_r^- G_r^+)/2$, $a_r = (L_r^+ b_r^- + L_r^- b_r^+)/2$ and $\psi_r = a_r - B_r\theta_0$. Conditional on the training data, $n^{-1}\sum EB_r \to B$ with B nonsingular, and $n^{-1}\sum \mathrm{Var}(\psi_r) \to V_\psi$ positive definite. A law of large numbers holds for B_r and the Lindeberg condition holds for ψ_r.

(A5) Exact integration is used, or the average instrumented quadrature defect is $o(n^{-1/2})$. This condition concerns centered continuous-parameter inference. A vanishing average defect alone suffices for consistency.

Theorem 3 (heterogeneous-record weak estimating equations). Under (A1)–(A5), conditional on the independent training data,

$$\widehat{\theta}_n = \left(\sum_{r=1}^{n} B_r\right)^{-1} \sum_{r=1}^{n} a_r \;\xrightarrow{P}\; \theta_0, \qquad \sqrt{n}(\widehat{\theta}_n - \theta_0) \;\Rightarrow\; N(0, B^{-1} V_\psi B^{-\top}).$$

The empirical covariance of the p-dimensional record scores, combined with the mean derivative $\bar{B}_n$, consistently estimates this sandwich. Thus a rich weak family requires a $p \times p$ empirical score covariance, while its within-record weights can come from the full model covariance in Proposition 3. Fixed-noise consistency applies to the stated independent-instrument construction. It does not follow merely by inserting a same-record pilot into $D_r^\top \Omega_r^{-1}$.

Proof. Conditional on u_r and training data, L_r^+ is independent of $m_r^-(\theta_0, \nu_0)$, and conversely. Hence

$$E(\psi_r \mid u_r, \text{training}) = \tfrac{1}{2} E L_r^+ \, E m_r^- + \tfrac{1}{2} E L_r^- \, E m_r^+ = 0.$$

The exact identity $\widehat{\theta}_n - \theta_0 = \bar{B}_n^{-1} \bar{\psi}_n$ gives consistency by the law of large numbers, followed by the displayed CLT by Lindeberg and Slutsky. The empirical score covariance converges to V_ψ: each conditional mean is zero, so variation of the deterministic trajectories creates no between-record mean term. Replacing θ_0 by the consistent estimate changes this covariance by $o_P(1)$ under the moment conditions. The quadrature assumption makes its centered contribution $o_P(1)$ after multiplication by $\sqrt{n}$. This is the standard estimating-equation/GMM argument [1, 2], with the independent-instrument identity addressing the incidental-initial-state setting. □

S16.2. Gaussian construction and local information reference

For known C_{ν_0}, draw $\zeta_r \sim N(0, C_{\nu_0})$ independently and set $Y_r^\pm = Y_r \pm \zeta_r$. Their cross-covariance is $C_{\nu_0} - C_{\nu_0} = 0$; jointly Gaussian views are independent, have identical mean and covariance $2C_{\nu_0}$. Together they recover the original record. This Gaussian randomization is an established data-fission construction [3]. It is used here to check finite-noise instruments, not proposed as a new general splitting method.

For a supplied clean trajectory, write $D_r = E G_r$ and $\Omega_r = \mathrm{Cov}(m_r(\theta_0, \nu_0))$. The deterministic optimal instrument within the prescribed linear moments is $L_r = D_r^\top \Omega_r^{-1}$. Direct calculation gives the oracle covariance

$$\left(\sum_r D_r^\top \Omega_r^{-1} D_r\right)^{-1}.$$

This is a fixed-family moment reference at finite noise. Proposition 3 evaluates Ω_r including all Hermite orders. A noisy $G_r^\top$ in place of $D_r^\top$ introduces $E(G_r^\top m_r)$ and can invalidate the zero-mean score. Independent instruments preserve the mean identity while generally changing the finite-noise covariance from this oracle expression.

In the local Gaussian experiment, each record has coefficient sensitivity S_r and initial sensitivity Z_r. The joint information for $\theta, u_1, \ldots, u_n$ has separate nuisance blocks. Its coefficient Schur complement is exactly

$$\mathcal{J}_{e,n} = \sum_{r=1}^{n} \left(\mathcal{J}_{\theta\theta}(u_r) - \mathcal{J}_{\theta u}(u_r)\mathcal{J}_{uu}(u_r)^{-1}\mathcal{J}_{u\theta}(u_r)\right) = \sum_{r} \mathcal{J}_{e}(u_r).$$

Theorem 2 applies record by record. With known shape and noise covariance $\epsilon^2 C$, a fixed test span gives the order-one moment information $D_r^\top \Omega_{r,1}^{-1} D_r$. Dense spans approach $\mathcal{J}_e(u_r)$ for any fixed finite collection of regular trajectories. Thus $(\sum_r \mathcal{J}_e(u_r))^{-1}$ is the sharp local reference for boundary-free influences in the product experiment. This exact block calculation alone supplies neither a fixed-noise incidental-parameter MLE theorem nor a growing-record, growing-span uniform efficiency result.

The symmetric independent-view score also preserves the leading influence under the following additional local conditions: its instruments converge in $L^{2+\delta}$ to the deterministic correct-covariance instruments; their perturbation is $O_P(\epsilon)$; weak moments have an L^2 expansion with $O(\epsilon^2)$ remainder; and covariance solves remain regular for the fixed span. Then

$$\tfrac{1}{2}\{L_r^+ m_r^- + L_r^- m_r^+\} = L_{r,0}\, m_{r,1}(\eta_r) + O_{L^2}(\epsilon^2).$$

Indeed the first-order moments are linear in $\eta_r \pm \zeta_r$, their average cancels ζ_r, and instrument perturbations multiply moments of order ϵ. The factor two in each view's covariance scales $L_{r,0}$ uniformly and cancels from the normalized estimating equation. Combined with Theorem 2, this proves the conditional local product-experiment efficiency statement for fixed n and sequential span enrichment. The fixed-noise CLT above and the local information reference describe complementary properties of the same weak scores; joint rates remain open.

S16.3. Covariance estimation and common-trajectory sufficiency

For an independently calibrated covariance estimate $\widehat{\nu}$, asymptotic linearity of the averaged score adds $H\xi$ to its influence, where $H = E\partial_\nu \psi$ and ξ is the calibration influence with the appropriate relative sample-size scaling. The Newey–McFadden nuisance argument then gives its sandwich [2]. For estimated-covariance Gaussian fission, calibration error also perturbs cross-view independence; its contribution must be included or be negligible at the inference scale. The known-covariance construction proves the stated result without presuming those extra conditions.

Uncorrected residual-profile estimates from separate fixed-horizon trajectory fits need not be consistent as records accumulate: each fit estimates a new initial state with nonvanishing error. Even in a linear Gaussian model, fitting a mean projects part of the error out of the residual and changes its expected quadratic form. A residual estimator therefore requires a demonstrated bias correction or a separate consistent calibration scheme. Pair differences eliminate the mean only for equal trajectories. These facts prevent transferring the common-trajectory pair estimator to heterogeneous initial states.

If all records instead share one initial state and known covariance, the Gaussian path average is sufficient for the mean parameter: the sample contrasts are parameter-free independent Gaussian observations. Applying a single-record procedure to the path average reduces covariance to C/n. For a nonlinear polynomial, a Hermite covariance term of order q then scales as n^{-q}, whereas averaging separate corrected moments retains its fixed-amplitude contribution divided by n. Path averaging is the natural reference in this common-trajectory experiment. The old fixed-family GMM calculation is retained as Lemma 2 and a calibration diagnostic, rather than as an efficiency contribution.

References

[1] Hansen LP. Large sample properties of generalized method of moments estimators. *Econometrica* 50 (1982) 1029–1054. https://doi.org/10.2307/1912775.

[2] Newey WK, McFadden D. Large sample estimation and hypothesis testing. *Handbook of econometrics* Elsevier 4 (1994) 2111–2245. https://doi.org/10.1016/S1573-4412(05)80005-4.

[3] Leiner J, Duan B, Wasserman L, Ramdas A. Data fission: Splitting a single data point. *Journal of the American Statistical Association* 120 (2025) 135–146. https://doi.org/10.1080/01621459.2023.2270748.